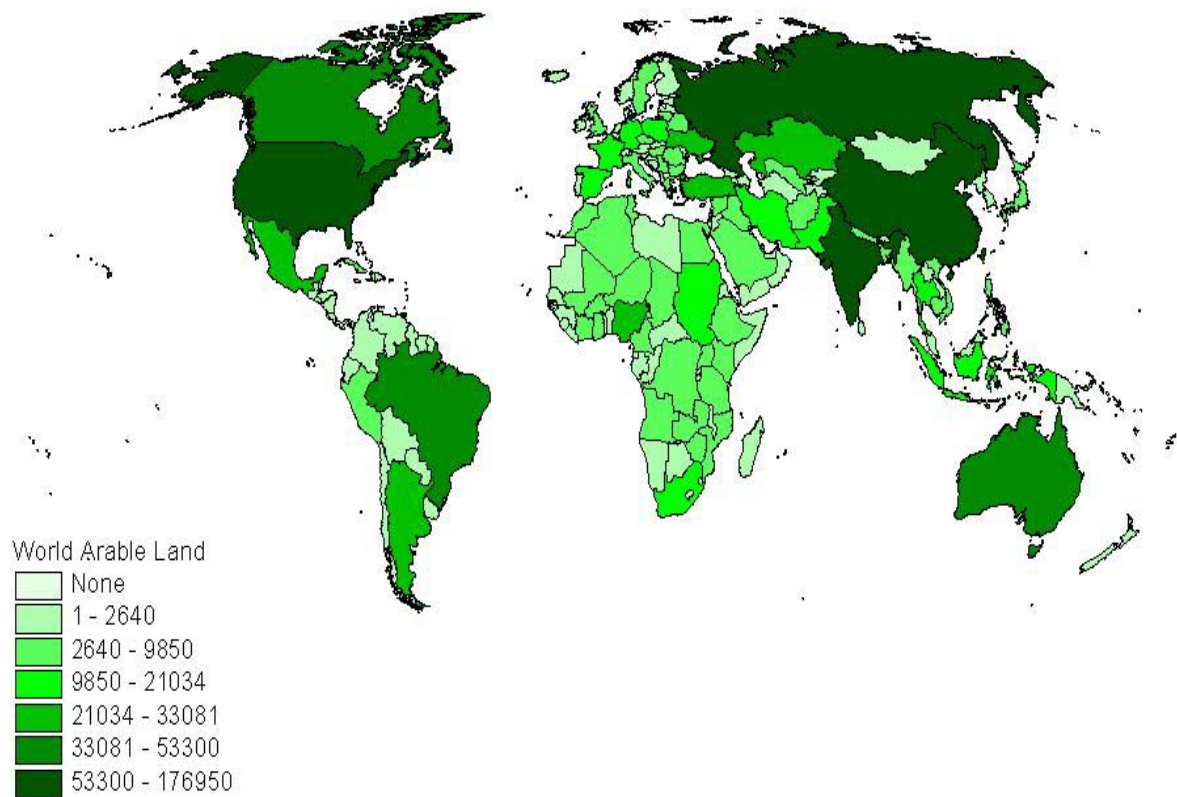




GLOBAL SUSTAINABLE AGRICULTURE

Sq. Km. of Arable Land



ARCVIEW LESSON

GIS Information

Here's a list and explanation of important features in ArcView:

Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

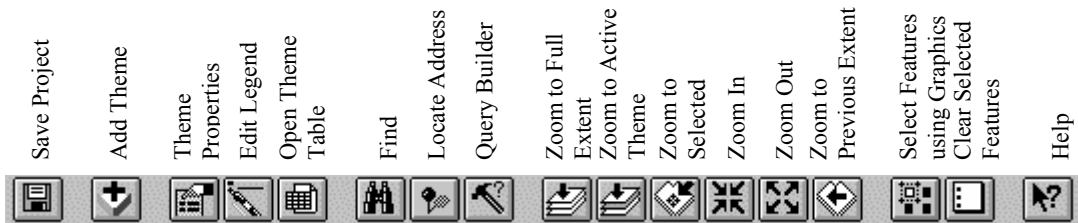
Tables  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

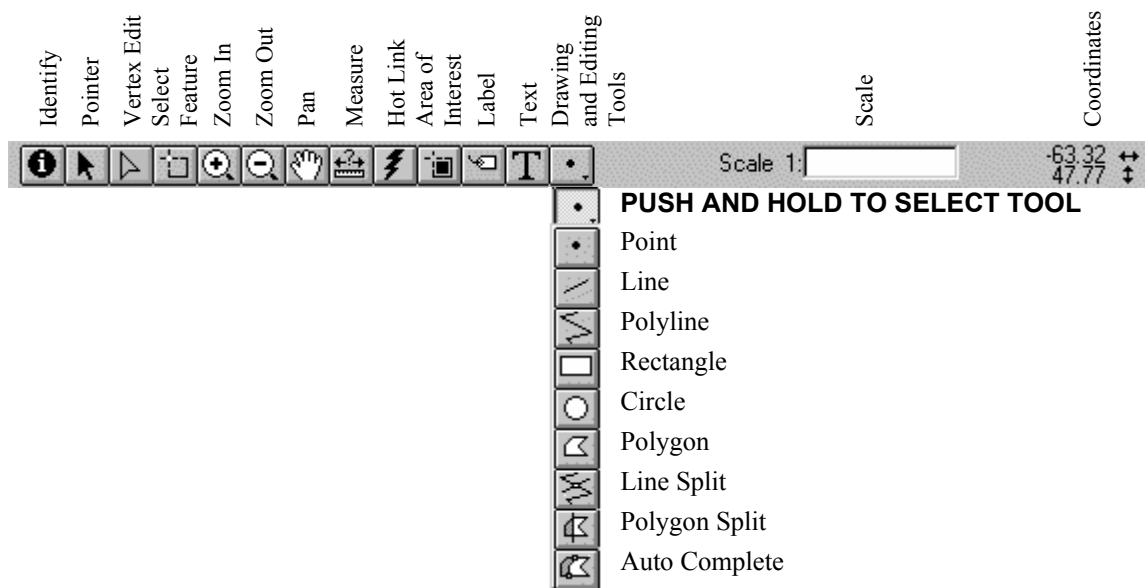


Table - Buttons

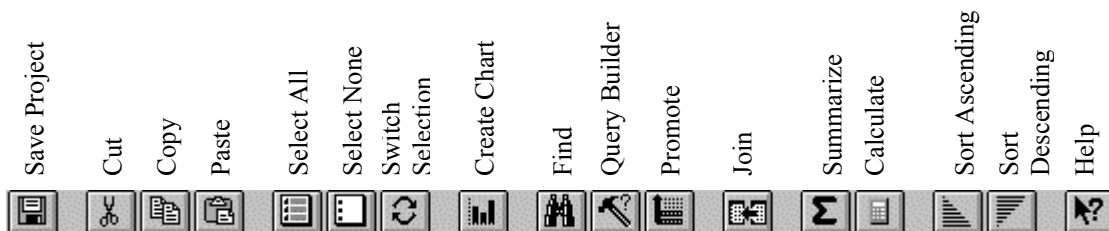


Table – Tools



Chart - Buttons

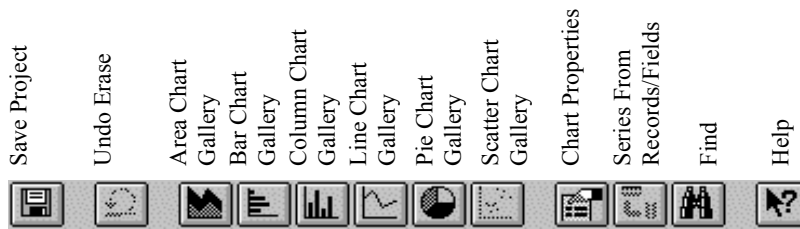
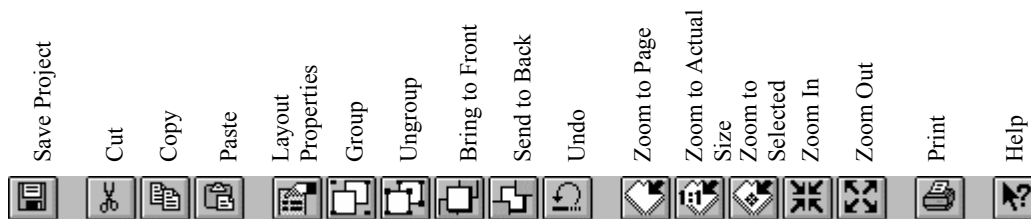


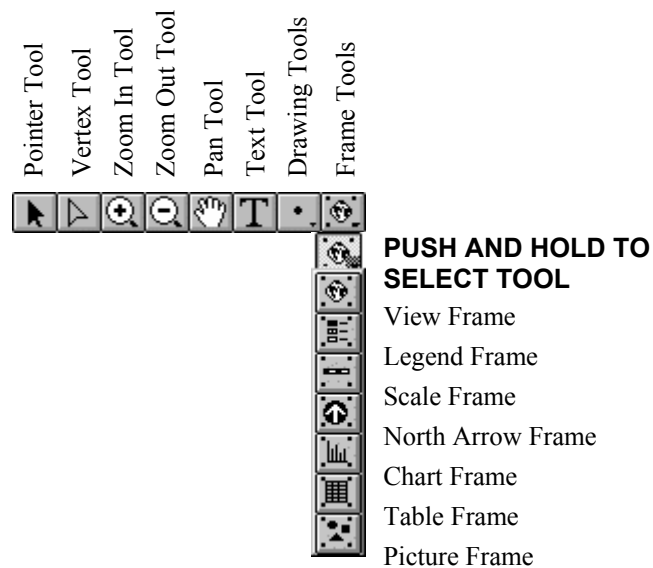
Chart - Tools



Layout - Buttons



Layout - Tools



For more Information: <http://www.esricanada.com>

Getting Started

Before you begin, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

! (The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).*

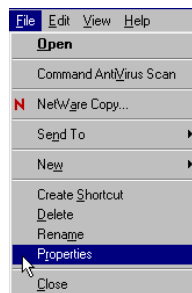
To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /**.

Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For windows based systems follow these instructions:

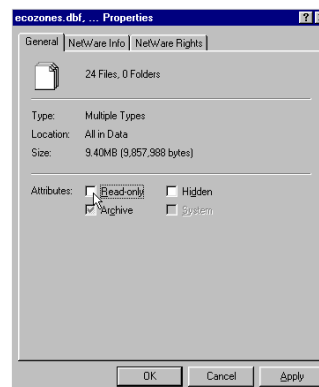
! (The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)

1. Open your new data folder
2. Choose **Edit > Select all** (they will all appear Blue)

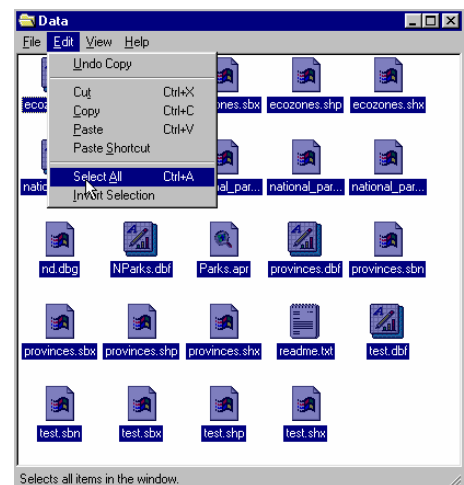
3. From the menu choose:
File > Properties



4. Remove the check mark in the box beside the label **read only**.



5. Click **apply**, then **OK**



The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

***!** Denotes an observation.

? Denotes a question to be answered on the Student Activity Worksheet 11.

STARTING THE PROJECT

The purpose of this lesson is to include new data in a table and computing new data within a table. The data will be displayed through charts, recoding polygons, and buildings queries and prepare a map layout for printing. The final step will be to prepare a map layout for printing. Global agriculture will be examined in terms of arable land, production, imports and exports, and fertilizer use.

STARTING ARCVIEW

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **O.K.**

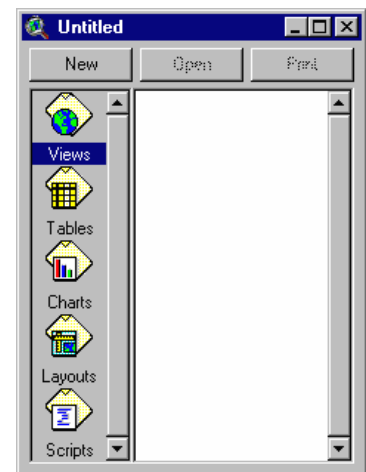


When the program starts, it automatically opens a project window titled "**Untitled.**"

If you like, you can maximize the **ArcView** window to fill the screen.

! In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.

3. Now you are ready to add documents to your new project.
4. With the "**Views**" icon selected, click on **New** and maximize the **View1** window.



5. Click the **Add theme button**  and Navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

population.shp
base.shp
world.shp

6. Click **O.K.**

This will add these three themes to the table of contents of the new view. They are now available for display and editing.



7. **Click and drag** to move the themes up or down in the table of contents to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line themes should be on top of polygon files to be visible.

8. **Click** on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by **Clicking** on other themes.

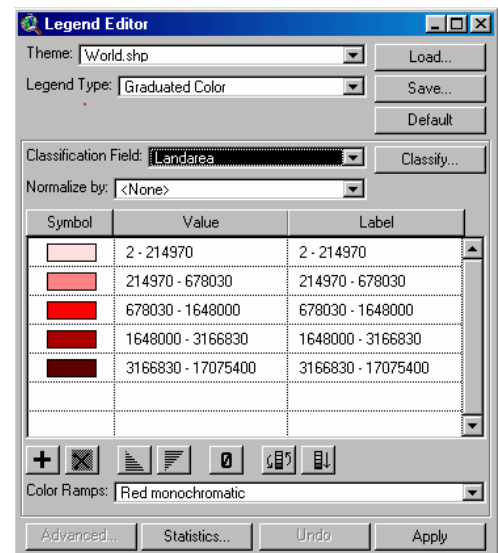
9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **apr.** means that you are saving your work as a project that can be opened up in the future)

ArcView - DISPLAYING A MAP

Mapping country size

1. From the **Project window** select the **Views** icon.
2. Select and open **View1**.
3. **Double Click** the **world.shp** theme to edit it.
4. Open **Legend Type** list.
5. Select **Graduated Colour**.
6. Open **Classification Field** list.
7. Select **Landarea**
8. Click **Apply**.
9. Close **Legend Editor** window.

! This will display the land area of each country in units of 1000 ha.



? What are the three largest countries in the world?

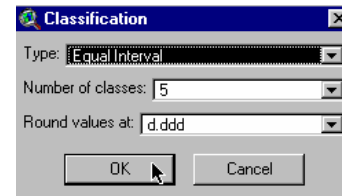
? What is the world's largest country and in which continent is it located?

Use the Identify  Tool to query countries to get their names and other details.

ArcView - CLASSIFYING A MAP

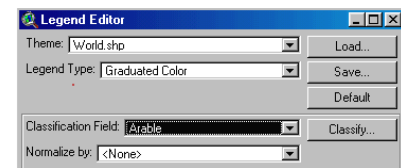
1. Double Click the *world* theme.
2. Click **Classify**.
3. Select **Type Equal Intervals**.
4. Click **O.K.**
5. Click **Apply**.
6. Close **Legend Editor** window.

? How did the map and legend change?



Repeat this process to show the area of arable land, (Arable).

1. Double Click the *world* theme.
2. Click **Classify**.
3. In the **Classification Field** list replace *landarea* with *Arable*.
4. Click **Apply**.
5. Close **Legend Editor** window.



? What are the differences between land area and arable land area?

? Describe the global distribution of arable land.

! If time permits try to map other information.

ARCVIEW - COMPUTING NEW DATA

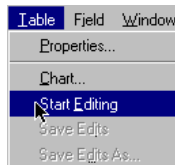
We will take data we have on land area and arable land area to calculate the percentage of arable land in each country.

1. Make the *world.shp* theme your active theme (it will appear raised).

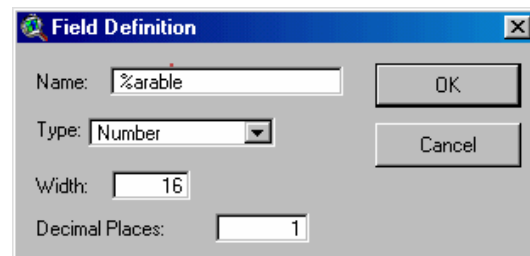
2. Click **open theme/ table** button.



3. From the menu select **table**.
4. Select **start editing**.



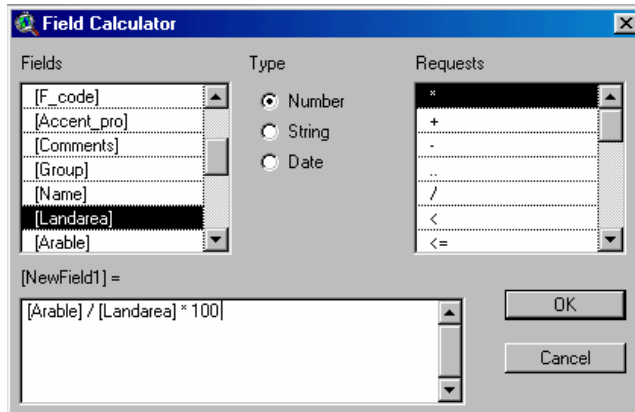
5. Click menu **edit**.
6. Select **add field**.
7. Change *name* to *%arable*.
8. Change **decimal place** to **1**.
9. Click **O.K.**



10. Click **calculate** button.

11. Build an **expression** to create the mathematical equation to calculate the percent of arable land area by selecting **field**, double click **Arable**, then choose **requests**, double click **/** double click **field** **landarea**, then choose **requests**, double click ***** then enter in the expression string 100. See the expression created. (arable / landarea * 100)

12. Click **O.K.**



13. Click menu **table**.

14. Select **stop editing**.

15. Click **yes** to save edits.

15. Close **table** window.

16. Use the mapping routine to map % **Arable** land.

? What parts of the world have the highest percentage of arable land?

ARCVIEW - DISPLAYING DATA WITH CHARTS

1. Double click **world** theme.

2. Open **legend type** list.

3. Select **chart**.

4. Click **landarea** field.

5. Click **add**.

6. Repeat for **Arable** field.

7. Click **pie chart** type.

8. Double click **background symbol** and change colour to a light shade by clicking on **paintbrush** and selecting a new colour.

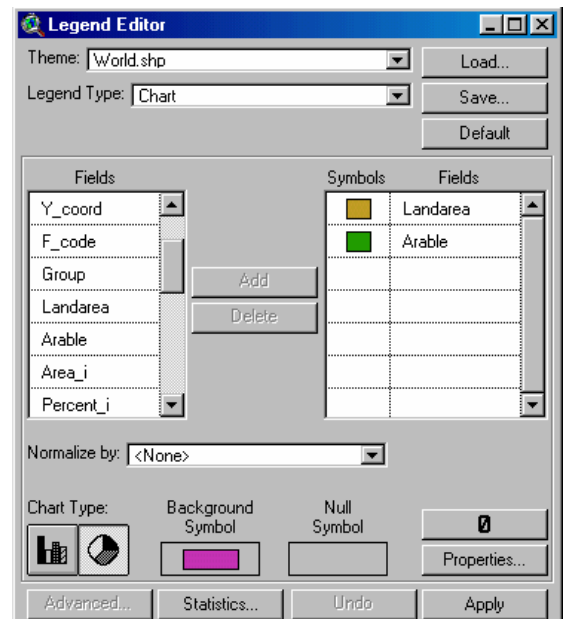
9. Click **properties** button.

10. Select **landarea** in the **Size Field**.

11. Click **O.K.**

12. Click **apply**.

13. Close **legend edit** window and **colour palette** window.



14. Click on **identify** button. 

! Note that only one chart is applied to each country, therefore countries with more than one polygon may have non-assigned areas.

? Find the names of five countries with a high percentage of arable land.

? Find the names of five countries with a small percentage of arable land.

ArcView - BUILDING A QUERY

1. **Double Click** the **world.shp** theme to edit it.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select **% Arable**.
8. Click **Apply**.
9. Close **Legend Editor** window.

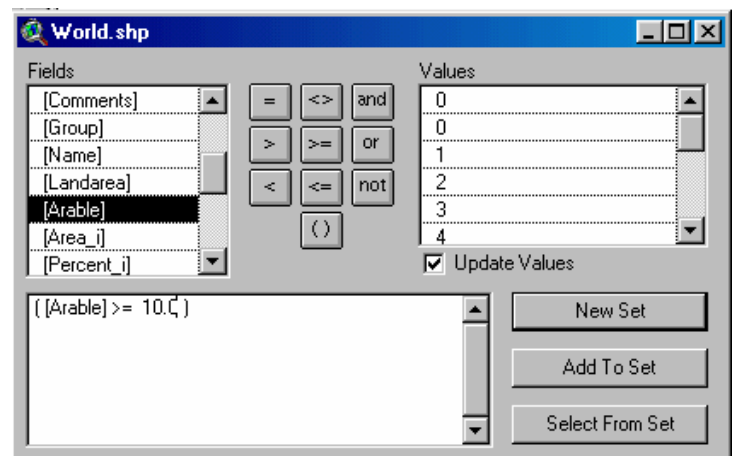
! This layer shows the percent of arable land for each country.

Now let's build a query to find all countries with more than 10% arable land.

1. Select and activate **world** theme.
2. Click **Query Builder** button. 
3. **Double Click Field %arable**.
4. Click **>=** button.
5. **Enter 10.0**.
6. Click **New Set** button.
7. Close **query window**.

? Where can you find countries with more than 10% arable land?

8. Click **Clear Selected Features** button.



! Note you must always remember to keep your total query equation contained in a set of brackets e.g. **[(arable)>= 10.0]**.

Using the same process as used for **%arable**, choose **Prod1998** (to display the countries where agricultural production was greater than 100% of 1989-91 production).

9. Click **Query Builder** button. 
10. **Double Click Field Prod1998**.
11. Click **>=**.
12. **Enter 100.0**.
13. Click **New Set** button.
14. Close **query window**.

? How many countries have had increases in agricultural production (use the table button)?

? What else does this map tell us about the location of countries with decreases in agricultural production?

Let's use this same process to query other aspects of the global agriculture. Data which is available.

1. Population density POPDEN (people per Sq. Km.)
2. Agricultural exports Expo1997 to Expo1970 (% of total exports)
3. Agricultural imports Impo1997 to Impo1970 (% of total imports)
4. Agricultural production Prod1998 to Prod1961 (% of 1989-91 production)
5. Agricultural labour force Labr1990 to Labr1960 (% of total labour force)
6. Fertilizer use Fert1999 to Fert1961 (thousand metric tons)

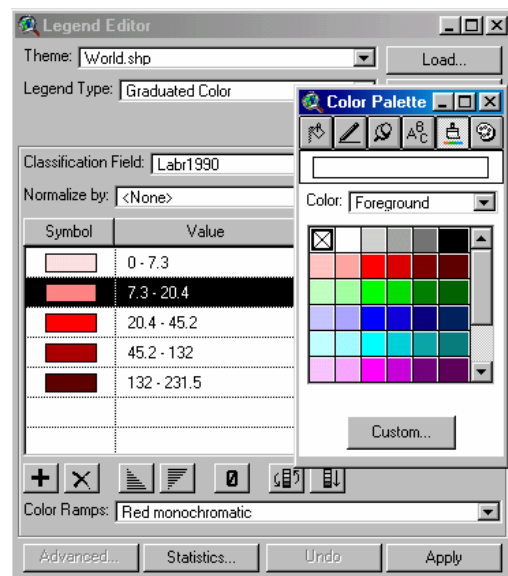
? Which Countries used over 2 000 000 000 metric tons of fertilizer in 1970 and 1985? (Hint: You can use the “and” button in the query builder to enter more than one field in a query)

ArcView - CHANGING DISPLAY FEATURES

Now that we know how to display a map, let's try different symbols and colours for the map units to make our map more meaningful.

1. Double Click the **world** theme.
2. Open **Legend Type** list.
3. Select **Graduated Color**.
4. Open **Classification Field** list. Select **Labr1990**.
5. Double Click symbol next to **the first unit**.
6. Click **Colour Brush** button.
7. Select a New Colour for the symbol representing that first unit.

! Note that the colour of the foreground, background. Outline and text can be changed to suit your map by using the **Color** window.



! Select an additional aspect of agriculture (e.g. Livestock, tractors, rice, etc.). Research the selected topic for 10 different countries to support a background paper and a GIS project.

Find, select and acquire data for the 10 countries to be used in the GIS project. Add the data to the world.dbf for GIS products.

(http://earthtrends.wri.org/searable_db/index.cfm?theme+10)

Define and Research and aspect of sustainable agriculture which can be illustrated with maps, graphs and tables using GIS. Define the implications for sustainable development in the 10 countries.

Prepare a report that presents your findings.

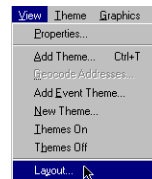
! You have completed your examination of Global Agriculture. Now create a layout for printing from any section of this analysis or create your own analysis for printing.

ArcView - Map Layout for printing

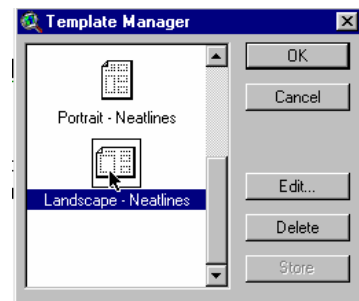
Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP

1. Click Menu **view**.
2. Click **Layout**.
3. Select the **layout template** you want to use
(One of the **landscape** templates is suggested).



4. Click **O.K.**
5. Maximize the **Layout** screen.



EDIT TITLE

1. **Double Click** on the title **View1**.
2. **Edit** the title to reflect the purpose of the map, e.g., forestry, population.
3. Click **O.K.**
4. **Resize** the title using the corner handles and move it into a position that is visually balanced.
5. **Click and Drag** to reposition the legend if necessary.

EDIT NORTH ARROW

1. **Double Click** on the *north arrow* and select a different presentation. Ensure that the arrow is pointing in the representative north direction of the map.
2. **Click and Drag** to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. **Double Click** the *scale bar*.
2. **Choose** a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. **Change** the units to **kms**.
4. Find a combination of style, scale and divisions that is visually pleasing
5. **Click and Drag** to position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. **Click** on the **Text** tool.
2. **Click** at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., “produced by (your name, date)”.
4. **Select** desired formatting information such as **alignment, spacing and rotation**.
5. **Click O.K.** when it appears, as you want it.
6. **Click** on **Pointer** tool.
7. **Select** the new text
8. **Click once** to change location or size, **Click twice** to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. **Click** menu **Layout**.
2. **Select Properties**.
3. **Change** the “name”. **Click O.K.**

ARRANGING YOUR LAYOUT

- !** Experiment with the layout buttons (use the tools page provided at the beginning of this lesson)
- to **group** (tie pieces together so they move as one)
 - to **ungroup** (break apart the ties so that individual pieces move)
 - **bring to front** (change order that pieces were laid down, e.g., lines on top of polygons)
 - **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
 - **zoom** (make image bigger or smaller)
 - **neat line** (border around layout)

PRINT MAP LAYOUT

1. **Click Print** button.
2. **Click Setup**.
3. **Select** your printer properties as outlined by your teacher.
4. **Click O.K.** to print.

! Note that the map can also be saved to a printer file.